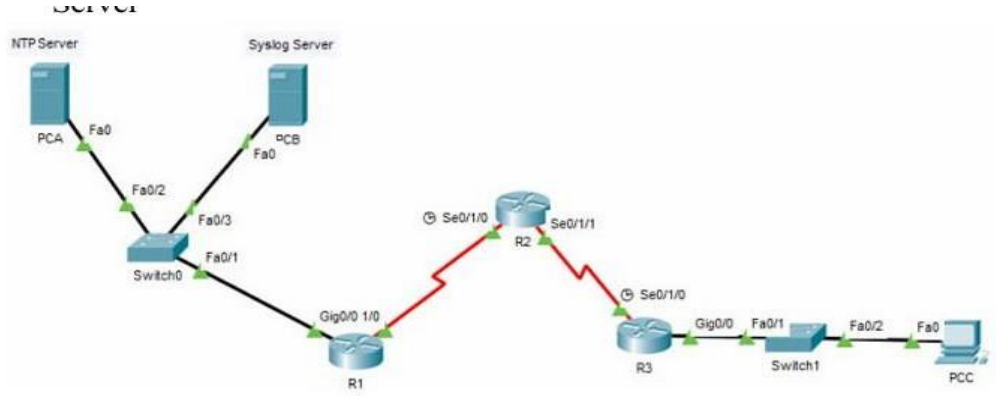
Practice Question

Q.1 Create the following topology and

♣ Configure OSPF MD5 authentication.

♣ Configure NTP and configure routers to log messages to the Syslog Server.

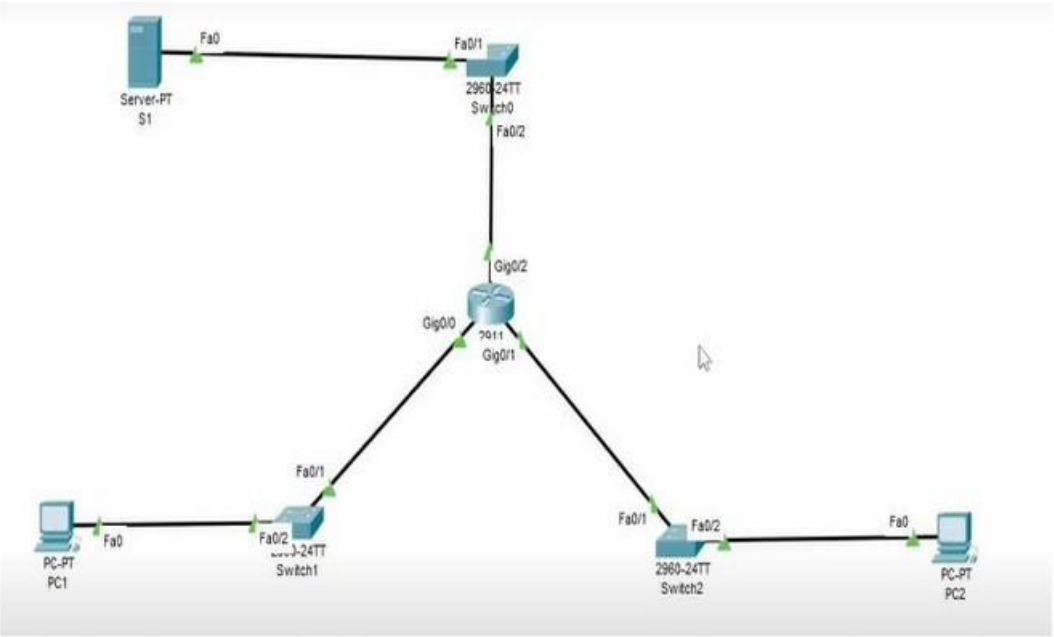


Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1

Q.2 Create the following topology and

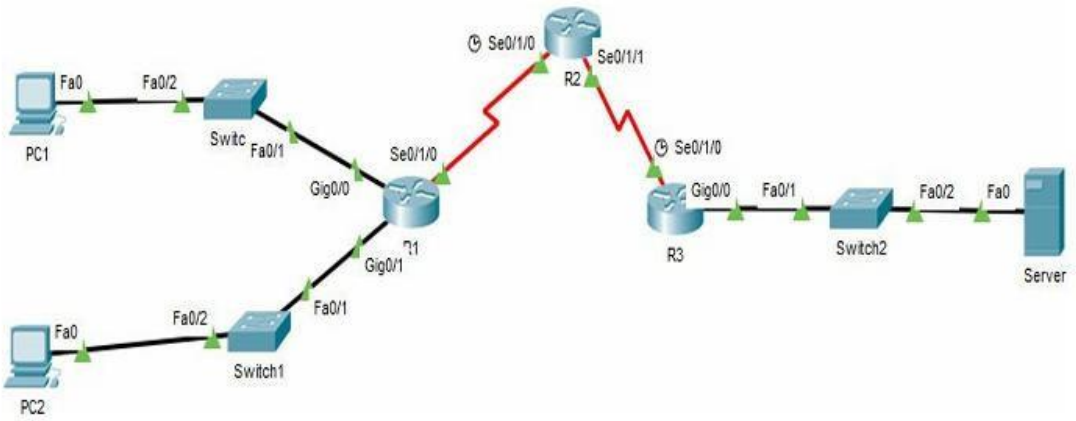
♣ Configure an ACL that will permit FTP and HTTP access on R1. ♣ Verify the ACL implementation. PC1 (Only FTP). PC2(Only HTTP)



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	172.22.34.65	255.255.255.224	N/A
	gig0/1	172.22.34.97	255.255.255.240	N/A
	gig0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

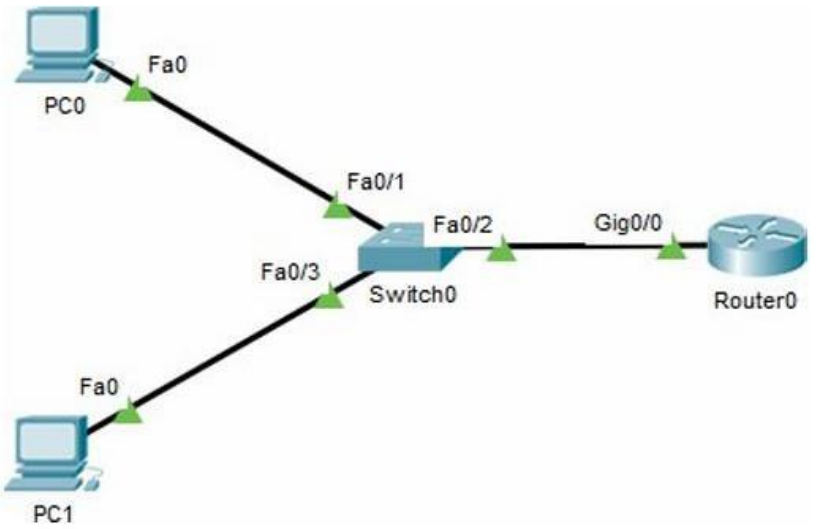
Q.3 Create the following topology using static routing and ♣ Configure, apply and verify an ACL that will block ICMP access on R3.



Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
PC1	NIC	2001:DB8:1:10::10/64	FE80::1
PC2	NIC	2001:DB8:1:11::11/64	FE80::1
R1	gig0/0	2001:DB8:1:10::1/64	FE80::1
	gig0/1	2001:DB8:1:11::1/64	FE80::1
	se0/1/0	2001:DB8:1:1::1/64	FE80::1
R2	se0/1/0	2001:DB8:1:1::2/64	FE80::2
	se0/1/1	2001:DB8:1:2::2/64	FE80::2
R3	gig0/0	2001:DB8:1:30::1/64	FE80::3
	se0/1/0	2001:DB8:1:2::1/64	FE80::3
Server	NIC	2001:DB8:1:30::30/64	FE80::3

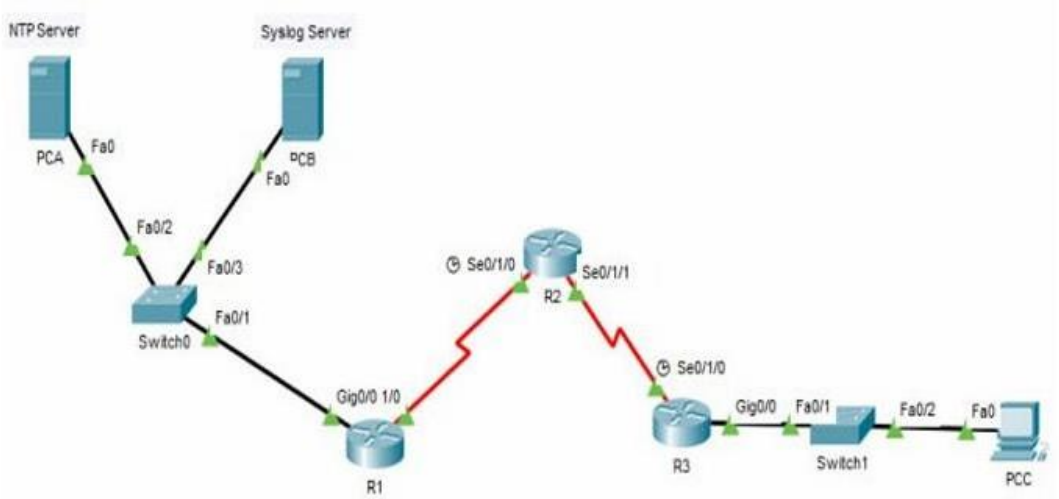
Q.4 Create the following topology and ♣ Configure OSPF MD5 authentication ♣ Configure a local user account on R1 and configure authenticate on the console and vty lines using local AAA. ♣ Verify local AAA authentication from the R1 console and the PC0 client and PC1 Client



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1

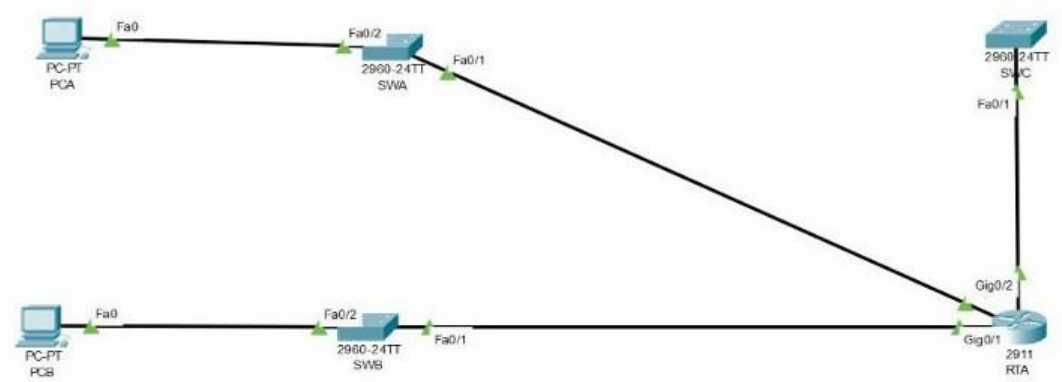
Q.5 **Create the following topology with OSPF routing and ♣ Configure NTP. ♣ Configure Routers to log messages to the syslog server. ♣ Configure R3 to support SSH connections.**



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1

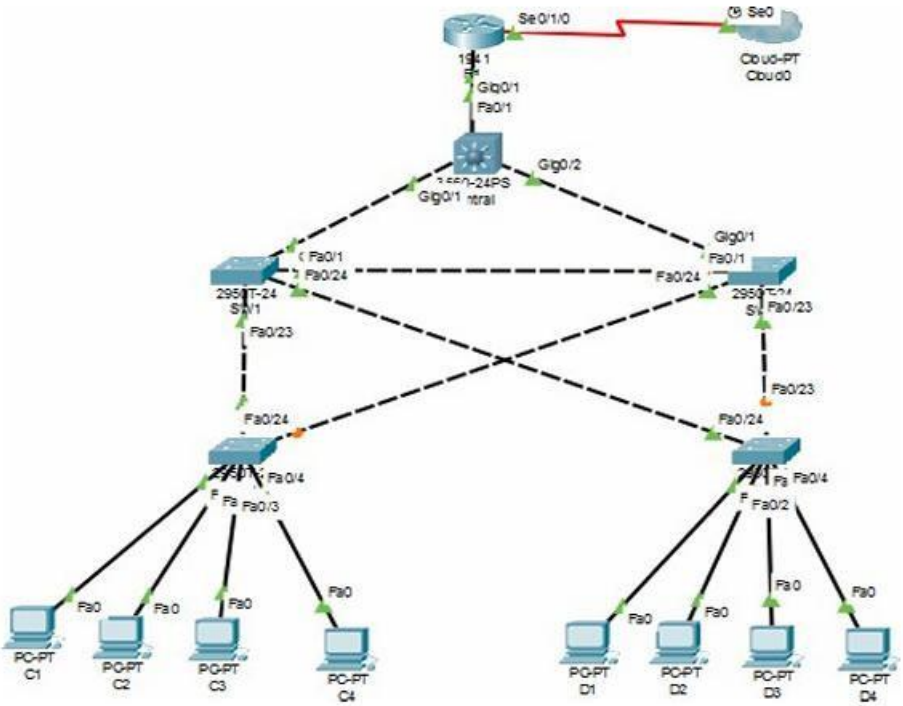
Q.6 Create the following topology and ♣ Configure an ACL that will permit one LAN to remotely access device in another LAN using SSH Protocol ♣ Besides ICMP all traffic from other network is denied. ♣ Verify the ACL implementation.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
RTA	gig0/0	10.101.117.49	255.255.255.248	N/A
	gig0/1	10.101.117.33	255.255.255.240	N/A
	gig0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN 1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN 1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN 1	10.101.117.2	255.255.255.224	10.101.117.1

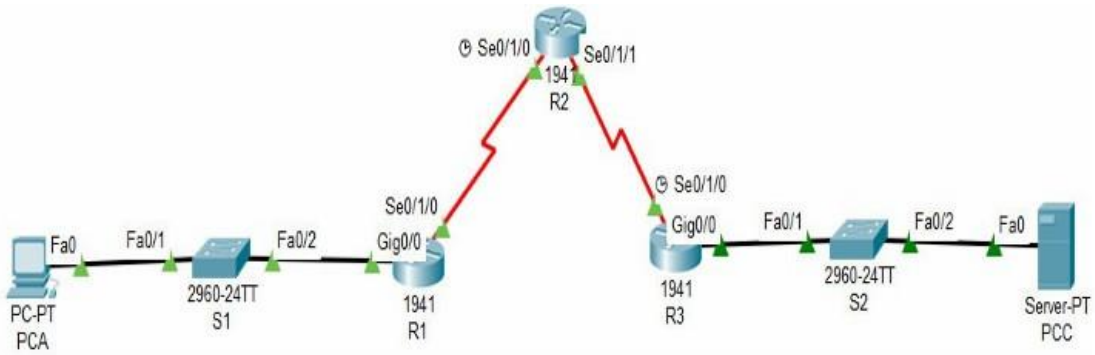
Q.7 Create the following topology and ♣ Assign the Central switch as the root bridge. ♣ Secure spanning-tree parameters to prevent STP manipulation attacks. ♣ Enable port security to prevent CAM table overflow attacks.



Addressing Table

Devices	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/1	192.168.1.1	255.255.255.0	N/A
	se0/0/0	209.165.200.1	255.255.255.0	N/A
C1	NIC	10.1.1.10	255.255.255.0	10.1.1.1
C2	NIC	10.1.1.11	255.255.255.0	10.1.1.1
C3	NIC	10.1.1.12	255.255.255.0	10.1.1.1
C4	NIC	10.1.1.13	255.255.255.0	10.1.1.1
D1	NIC	10.1.1.14	255.255.255.0	10.1.1.1
D2	NIC	10.1.1.15	255.255.255.0	10.1.1.1
D3	NIC	10.1.1.16	255.255.255.0	10.1.1.1
D4	NIC	10.1.1.17	255.255.255.0	10.1.1.1

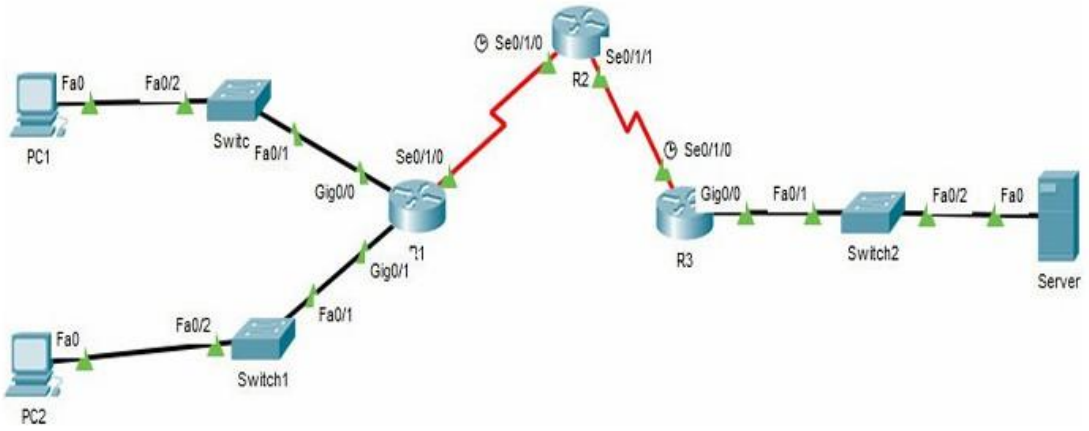
Q.8 Create the following topology using static routing and configure ♣ A zone-based policy (ZPF) firewall on R1 ♣ Verify ZPF firewall functionality using ping, SSH and a web browser.



Addressing table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

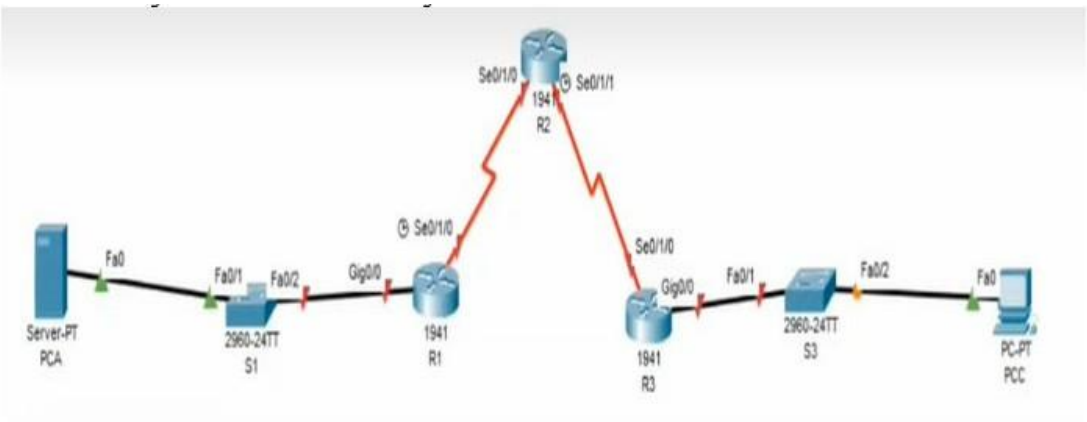
Q.9 Create the following topology using static routing and ♣ Configure, apply and verify an ACL that will block HTTP access on R1 ♣ Configure, apply and verify an ACL that will block HTTPS access on R1.



Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
PC1	NIC	2001:DB8:1:10::10/64	FE80::1
PC2	NIC	2001:DB8:1:11::11/64	FE80::1
R1	gig0/0	2001:DB8:1:10::1/64	FE80::1
	gig0/1	2001:DB8:1:11::1/64	FE80::1
	se0/1/0	2001:DB8:1:1::1/64	FE80::1
R2	se0/1/0	2001:DB8:1:1::2/64	FE80::2
	se0/1/1	2001:DB8:1:2::2/64	FE80::2
R3	gig0/0	2001:DB8:1:30::1/64	FE80::3
	se0/1/0	2001:DB8:1:2::1/64	FE80::3
Server	NIC	2001:DB8:1:30::30/64	FE80::3

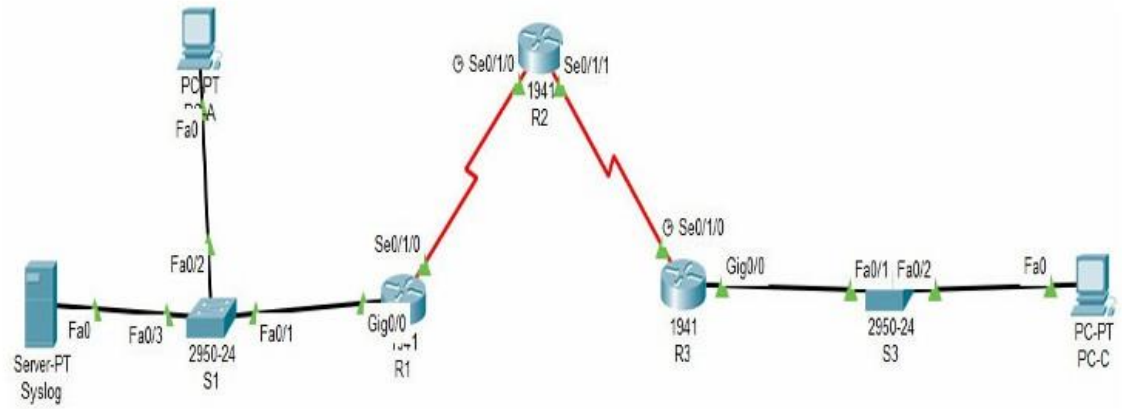
Q.10 Create the following topology using static routing ♣ Configure ACL to allow access to routers R1, R2, and R3 to only be permitted from PC-C, the management station. PC- C is also used for connectivity testing to PC-A, which is a server providing DNS, SMTP, FTP, and HTTPS services. ♣ Verify ACL functionality



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
	lo0	192.168.2.1	255.255.255.0	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

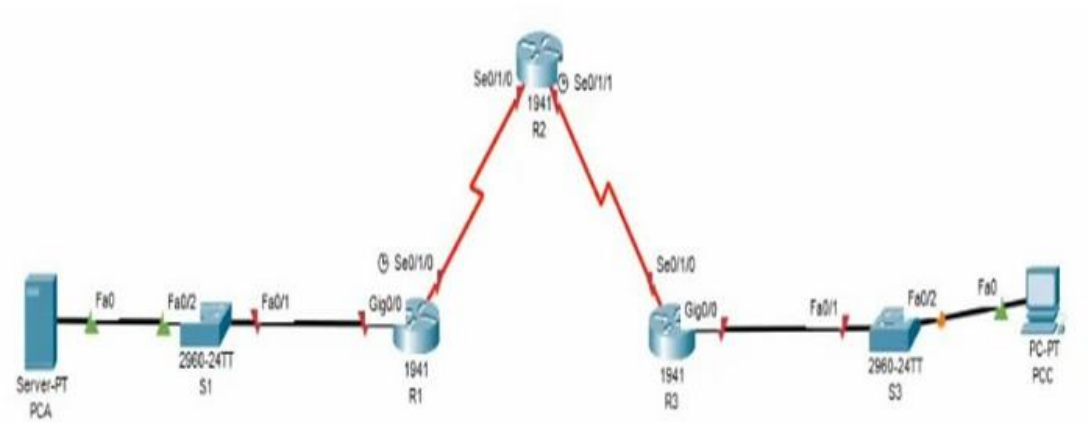
Q.11 Create the following topology and ♣ Enable IOS IPS ♣ Configure logging and verify IPS ♣ Modify signature and verify again.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1

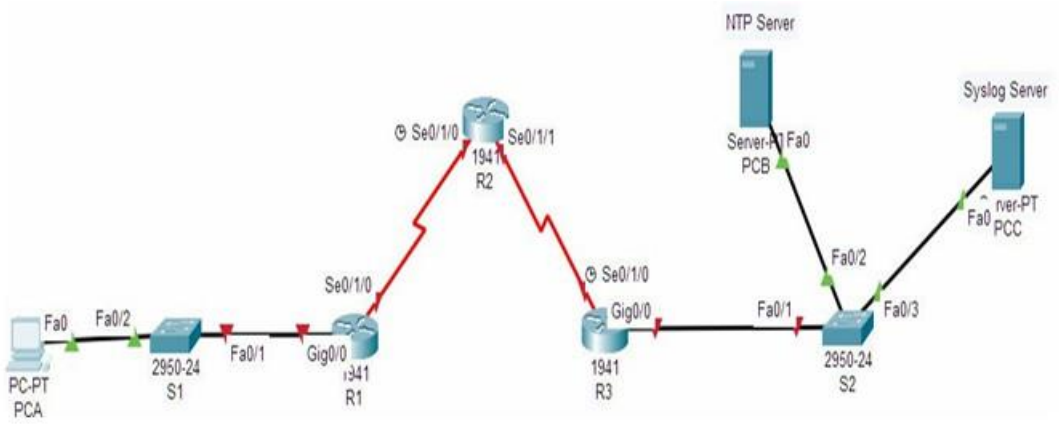
Q.12 Create the following topology using static routing and configure ♣ A zone-based policy (ZPF) firewall on R3 ♣ Verify ZPF firewall functionality using ping, SSH and a web browser.



Addressing table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

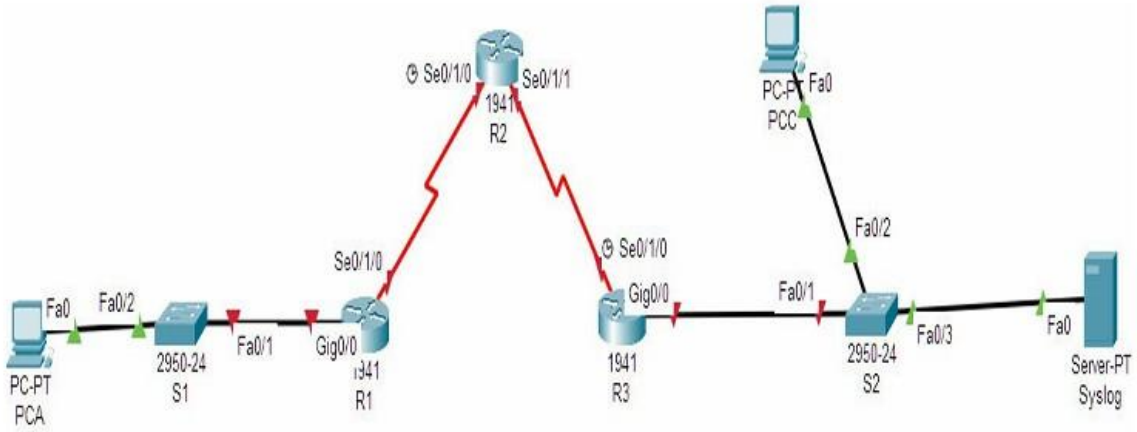
Q.13 Create the following topology and ♣ Configure OSPF MD5 authentication. ♣ Configure NTP and configure routers to log messages to the Syslog Server.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.3.5	255.255.255.0	192.168.3.1
PC-C	NIC	192.168.3.6	255.255.255.0	192.168.3.1

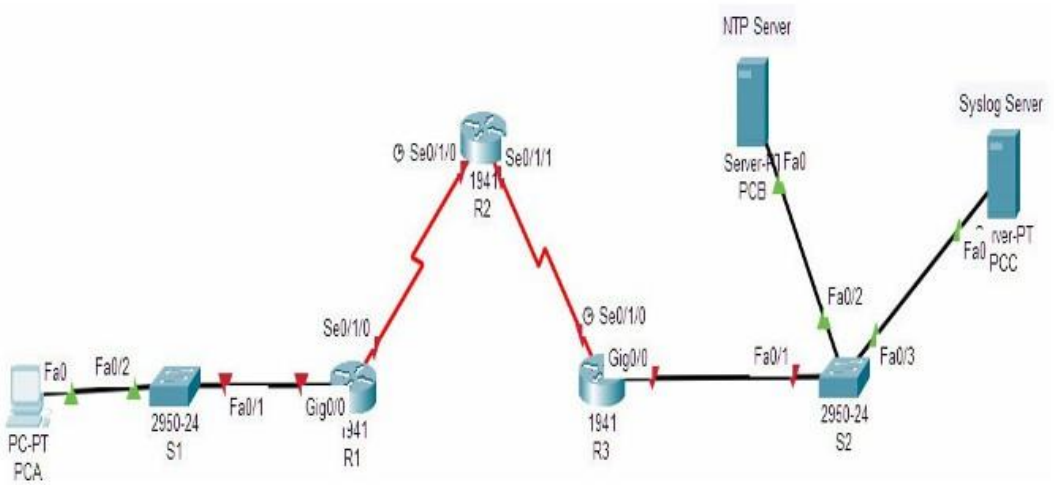
Q.14 Create the following topology and ♣ Enable IOS IPS ♣ Configure logging and verify IPS ♣ Modify signature and verify again.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
Syslog	NIC	192.168.3.50	255.255.255.0	192.168.3.1
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1

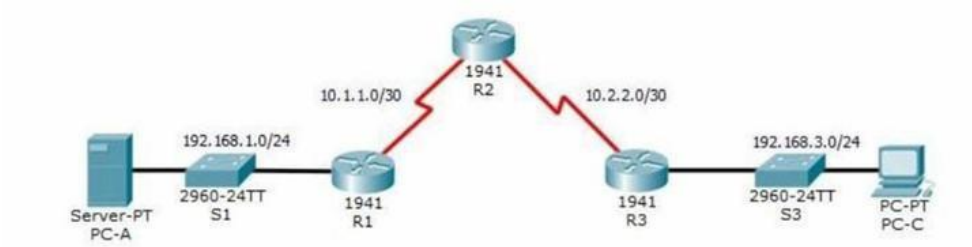
Q,17 Create the following topology with OSPF routing and ♣ Configure NTP. ♣ Configure Routers to log messages to the syslog server. ♣ Configure R1 to support SSH connections.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.3.5	255.255.255.0	192.168.3.1
PC-C	NIC	192.168.3.6	255.255.255.0	192.168.3.1

Q.18 Create the following topology to Configure IP ACLs to Mitigate Attacks.

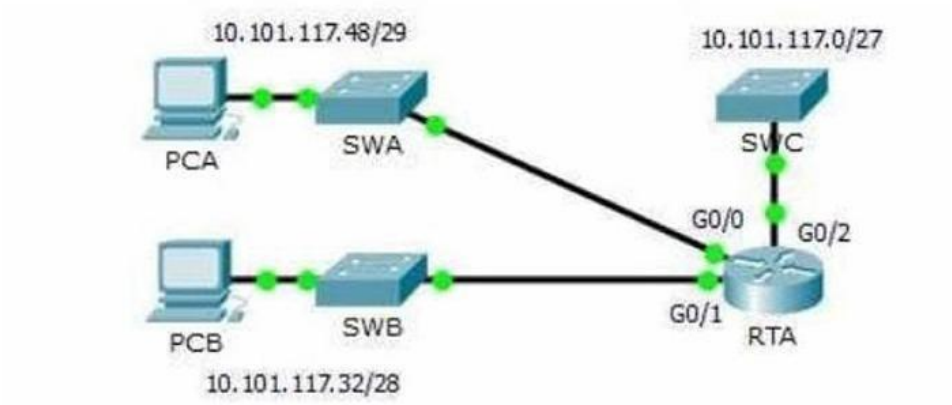


Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
	Lo0	192.168.2.1	255.255.255.0	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Q.19 Create the following topology using static routing

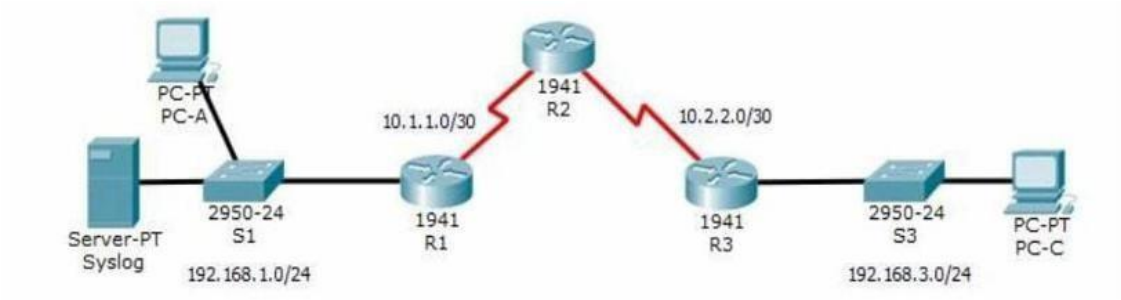
1. Configure the extended ACL
2. Verify the extended ACL implementation



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
RTA	G0/0	10.101.117.49	255.255.255.248	N/A
	G0/1	10.101.117.33	255.255.255.240	N/A
	G0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN 1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN 1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN 1	10.101.117.2	255.255.255.224	10.101.117.1

Q.20 Create the following topology to Configure IOS Intrusion Prevention System (IPS) Using the CLI.

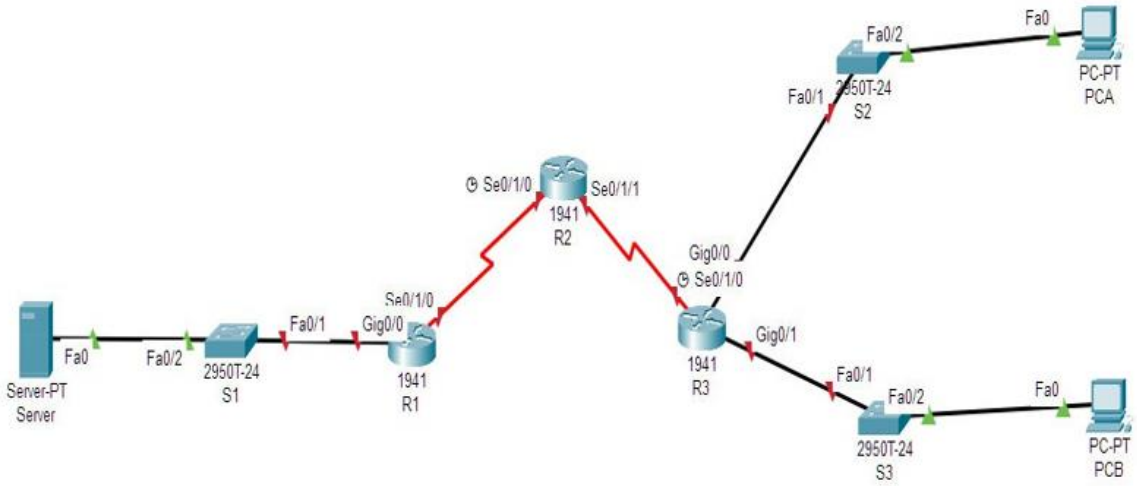


Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/1
	S0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1	S1 F0/2
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F0/3
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/2

Q.21 Create the following topology using static routing and

- Configure, apply and verify an ACL that will block HTTP access on R3
- Configure, apply and verify an ACL that will block HTTPS access on R3.



Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
Server	NIC	2001:DB8:1:10::30/64	FE80::1
R1	gig0/0	2001:DB8:1:10::1/64	FE80::1
	se0/1/0	2001:DB8:1:1::1/64	FE80::1
R2	se0/1/0	2001:DB8:1:1::2/64	FE80::2
	se0/1/1	2001:DB8:1:2::2/64	FE80::2
R3	gig0/0	2001:DB8:1:30::1/64	FE80::3
	gig0/1	2001:DB8:1:31::1/64	FE80::3
	se0/1/0	2001:DB8:1:2::1/64	FE80::3
PCA	NIC	2001:DB8:1:30::10/64	FE80::3
PCB	NIC	2001:DB8:1:31::11/64	FE80::3